

Lab 1, Act 1: Diagnose

How does predation risk depend on tadpole size?

NRES 746 – name: _____ group: _____

The scenario

Vonesh and Bolker (2005) studied *Hyperolius spinigularis* tadpoles in Tanzania, which face a trade-off: hatching earlier means hatching smaller, and smaller tadpoles may be more vulnerable to predation by dragonfly larvae – or they may be small enough to be ignored. To find out, they ran a field experiment: tadpoles of five different sizes were placed into tanks with a dragonfly predator, and after three days the researchers counted how many of the 10 tadpoles in each tank had been killed.

The data below are real measurements from that experiment. Today we'll use them to practice the math and data-exploration tools you'll need throughout the semester – starting with the most basic question a data analyst always asks first: *what does the pattern look like?*

Table 1: Vonesh & Bolker (2005) size-predation data

Tadpole size, TBL (mm)	Number killed (out of 10)
9	0
9	2
9	1
12	3
12	4
12	5
21	0
21	0
21	0
25	0
25	1
25	0
37	0
37	0
37	0

Part 1: Diagnose numerically (individual, ~8 min)

It's tempting to jump straight to a plot, but real data analysis usually starts with the numbers. Below is the mean and variance of “number killed” for each of the five tank sizes (across the 3 replicate tanks per size).

Table 2: Summary statistics by tadpole size

TBL (mm)	Kill values (3 tanks)	Mean	Variance
9	0, 2, 1	1.00	1.00
12	3, 4, 5	4.00	1.00

TBL (mm)	Kill values (3 tanks)	Mean	Variance
21	0, 0, 0	0.00	0.00
25	0, 1, 0	0.33	0.33
37	0, 0, 0	0.00	0.00

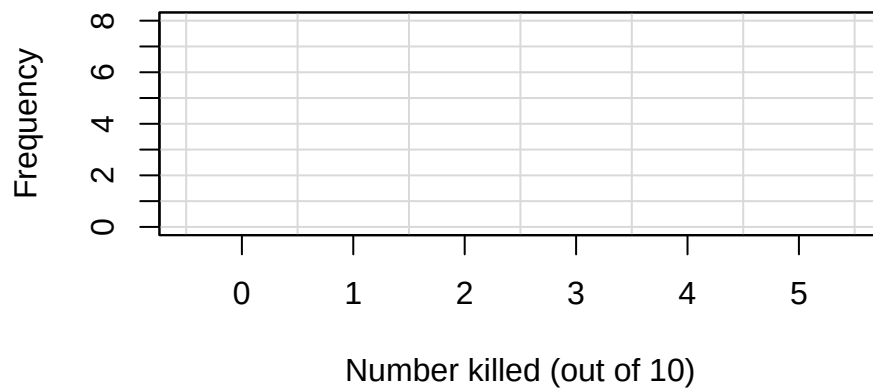
1a. Spot-check. Pick *one* row above and verify its mean and variance by hand below, just to confirm where these numbers come from. (Reminder: $\bar{x} = \frac{1}{n} \sum x_i$, $s^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2$.)

Row I'm checking: TBL = _____

1b. Does body size (TBL) appear to matter for predation risk? Describe the pattern in the means across the five sizes.

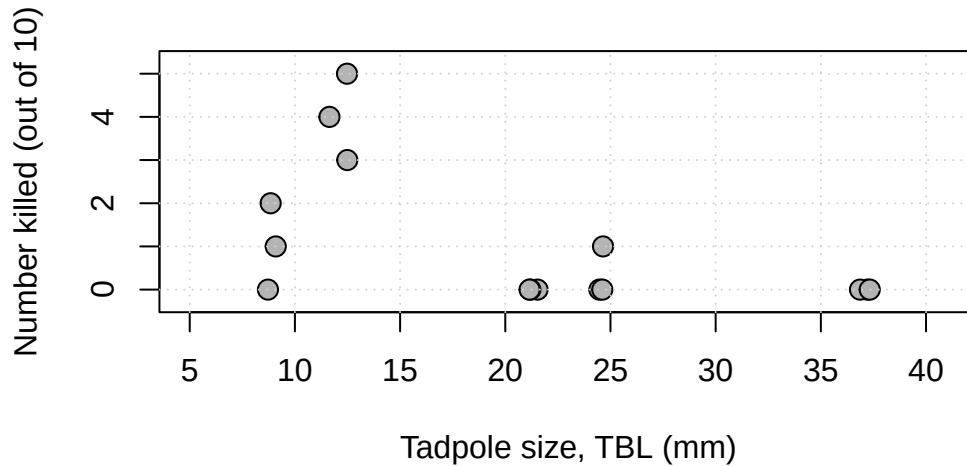
1c. Now compare the variances. Are they roughly similar across groups, or do some vary a lot more than others? Do you notice any relationship between a group's mean and its variance? Any guesses why some groups have variance = 0?

1d. Sketch a histogram. Using the full set of 15 individual tank counts (not the group means), sketch a histogram of “number killed” below, then describe its shape in a sentence or two.



Part 2: Diagnose visually (individual, ~4 min)

Below is a scatterplot of all 15 tanks: tadpole size (TBL) on the x-axis, number killed on the y-axis. (Points are jittered slightly left-right so overlapping tanks are visible.)



Sketch a smooth curve by hand directly on the plot above, showing what you think the underlying trend looks like (ignore tank-to-tank noise – draw the *pattern*, not a connect-the-dots line).

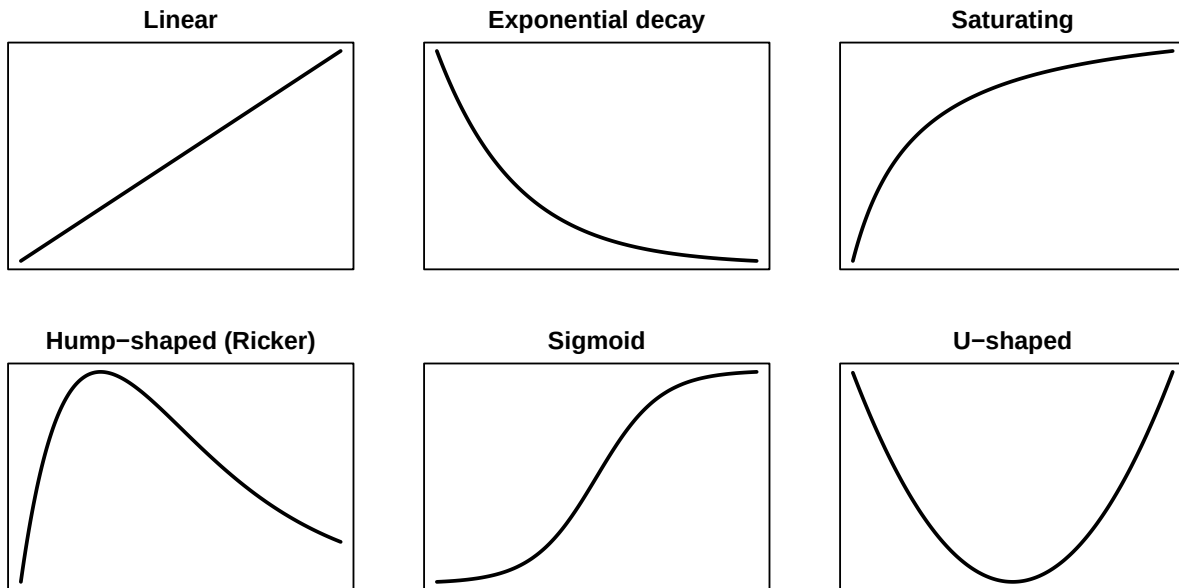
Classify the pattern. Circle whichever of these best describes what you drew:

monotonic increasing | monotonic decreasing | saturating (rises then flattens) | hump-shaped
(rises then falls) | roughly flat / no clear trend

In 1-2 sentences, what *ecological* story would explain the pattern you circled? (Hint: think about what small vs. large tadpoles might be able to do that changes their risk.)

Part 3: Name some candidates

We haven't learned the equations for these yet (that's Lab 2!) – but you've almost certainly seen shapes like these before. Here's a quick reference gallery of function shapes that come up constantly in ecology:



Based on the shape you circled in Part 2, which **one or two** of these gallery shapes seem like plausible candidates for the tadpole data? List them, and briefly say why you ruled out the others:

Group discussion (15 min)

Compare your answers with your group:

- Compare which row each of you spot-checked in Part 1 – between your group, how many of the 5 rows did you collectively verify, and did your numbers agree?
- Did everyone read the same pattern in the means and variances? Resolve any disagreements about what the numbers are telling you.
- Did everyone classify the pattern the same way in Part 2? If not, what in the data is each person pointing to?
- As a group, agree on your top candidate shape(s) from Part 3. What would it take to convince a skeptic that this shape – and not one of the others – is the right one?

Revise and reflect (individual, ~15 min)

Revise any of your answers above based on the group discussion (it's fine to cross out and rewrite – I want to see your thinking evolve). Then answer:

1. What is your group's best candidate function shape for this pattern, and what one feature of the data is the strongest evidence for it?

2. What would you still need to know (mathematically) to go from “this looks hump-shaped” to an actual equation you could fit to the data? (No wrong answers – “I don’t know” is a fine answer here.)

3. Muddiest point: what’s the one thing from this part of the lab you’d like to see explained again?
